

SciPy

What is SciPy?

- Open source python library, created in 2001
- Provides numerical algorithms for scientists and engineers
- Includes optimization, root finding, integration, and interpolation
- Makes proven numerical methods accessible without implementing them from scratch

Who uses SciPy and Why?

StakeHolders

- Researchers
- Engineers
- Students
- Software developers

What matters to them?

- Accurate and reproducible results
- Fast reliable computations
- Accessible numerical algorithms
- Maintained code and stable interfaces

How the community works together?

- Contributions from volunteers, researchers, and institutional developers
- Bugs and improvements discussed through GitHub issues and pull requests
- Significant changes discussed on the Scientific Python forum
- Changes reviewed by developers before merging
- Supported by a steering council, sponsors, and community teams

Libraries SciPy uses

- SciPy uses python as its main interface
  - How users call numerical functions
- SciPy numerical algorithms
  - Optimization, linear algebra, root finding etc.
- NumPy
  - Array operation and numerical data
- Compiled libraries
  - C/C++/Fortran and BLAS/LAPACK for relevant routines

How SciPy relates to Numerical Comp

SciPy connects directly to our class topics because it provides numerical methods for solving equations. Different methods trade off convergence, accuracy, computational cost, and stability.

EX:

$f(x) = x^3 - 2 = 0$

GOAL: find $x \approx 1.2599$

Method
- Bisection
- Newton's method

Strengths
- Reliable
- Very fast

Weakness
- Slower
- Sensitive to initial guess (x_0).

Takeaways
- SciPy provides multiple algorithms so users can pick the best method for their numerical problem.

Group project & Open Q’s

Possible Experiment

Comparing SciPy Root-Finding Algorithms

- Compare Bisection, Newton's method, and Brent's method.
- Change initial guesses and requested accuracy.
- Measure iterations, function evaluations, and convergence success.

Proposed graph

X-axis: Requested accuracy (error tolerance)

Y-axis: Number of function evaluations

Question for the Community

How does SciPy decide which root-finding methods to recommend for different problems?